INCH-POUND
MIL-PRF-123/16
21 November 1989

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PERFORMANCE SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC, (TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY, LEADED, STYLE CKS22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-123.

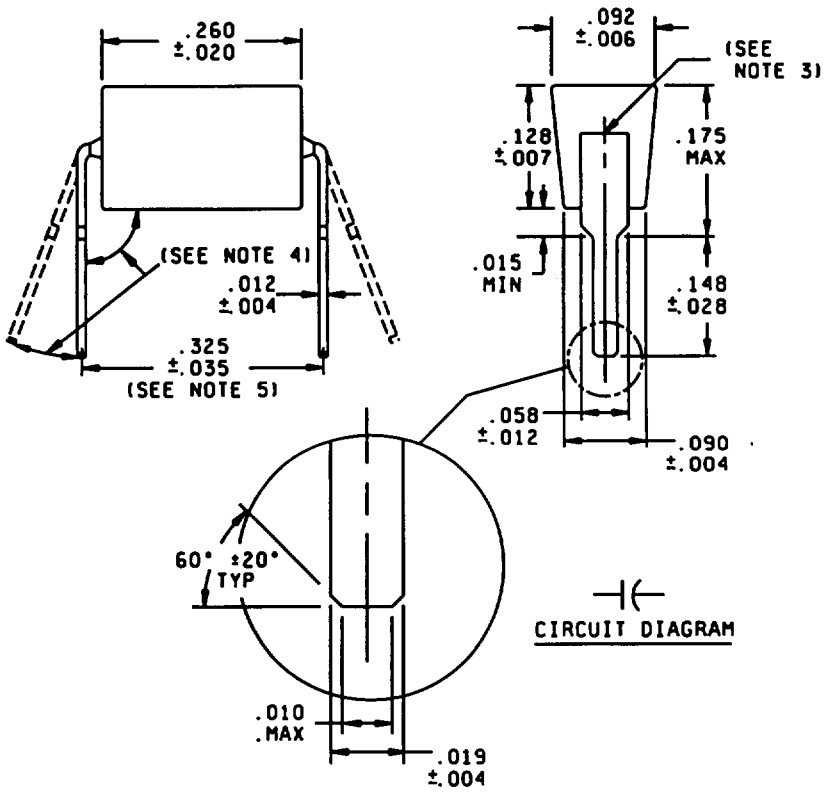



FIGURE 1. Style CKS22 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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| Inches | mm | Inches | mm |
|---|---|---|---|
| .004 | 0.10 | .035 | 0.89 |
| .006 | 0.15 | .058 | 1.47 |
| .007 | 0.18 | .090 | 2.29 |
| .010 | 0.25 | .092 | 2.34 |
| .012 | 0.30 | .128 | 3.25 |
| .015 | 0.38 | .148 | 3.76 |
| .019 | 0.48 | .175 | 4.44 |
| .020 | 0.51 | .260 | 6.60 |
| .028 | 0.71 | .325 | 8.26 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be centered within ±.005 (0.13 mm).
4. The angle shall be 95° +10°, -5°.
5. The distance between the centers of the mounting holes will be .300 ±.010 inch (7.62 ±0.25 mm).
6. Nonconductive material shall not extend beyond .030 inch (0.76 mm) from the edge of the capacitor body.

FIGURE 1. Style CKS22 capacitors - Continued.



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TABLE I. Style CKS22 characteristics.

| Part number 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (V dc) |
|---|---|---|---|---|
| M123A16BPD1R0DC | 1.0 | D | BP | 200 |
| M123A16BPD1R2DC | 1.2 | " | " | " |
| M123A16BPD1R5DC | 1.5 | " | " | " |
| M123A16BPD1R8DC | 1.8 | " | " | " |
| M123A16BPD2R2DC | 2.2 | " | " | " |
| M123A16BPD2R7DC | 2.7 | " | " | " |
| M123A16BPD3R3DC | 3.3 | " | " | " |
| M123A16BPD3R9DC | 3.9 | " | " | " |
| M123A16BPD4R7DC | 4.7 | " | " | " |
| M123A16BPD5R6DC | 5.6 | " | " | " |
| M123A16BPD6R8DC | 6.8 | " | " | " |
| M123A16BPD8R2DC | 8.2 | " | " | " |
| M123A16BPD100-C | 10 | D,J,K | " | " |
| M123A16BPD120-C | 12 | " | " | " |
| M123A16BPD150-C | 15 | " | " | " |
| M123A16BPD180-C | 18 | " | " | " |
| M123A16BPD220-C | 22 | " | " | " |
| M123A16BPD270-C | 27 | " | " | " |
| M123A16BPD330-C | 33 | " | " | " |
| M123A16BPD390-C | 39 | " | " | " |
| M123A16BPD470-C | 47 | " | " | " |
| M123A16BPD560-C | 56 | " | " | " |
| M123A16BPD680-C | 68 | F,J,K | " | " |
| M123A16BPD820-C | 82 | " | " | " |
| M123A16BPD101-C | 100 | " | " | " |
| M123A16BPD121-C | 120 | " | " | " |
| M123A16BPD151-C | 150 | " | " | " |
| M123A16BPD181-C | 180 | " | " | " |
| M123A16BPD221-C | 220 | " | " | " |
| M123A16BPD271-C | 270 | " | " | " |
| M123A16BPD331-C | 330 | " | " | " |
| M123A16BPD391-C | 390 | " | " | " |
| M123A16BPD471-C | 470 | " | " | " |
| M123A16BPC561-C | 560 | " | " | 100 |
| M123A16BPC681-C | 680 | " | " | " |
| M123A16BPC821-C | 820 | " | " | " |
| M123A16BPC102-C | 1000 | " | " | " |
| M123A16BPC122-C | 1200 | " | " | " |
| M123A16BPC152-C | 1500 | " | " | " |
| M123A16BPC182-C | 1800 | " | " | " |
| M123A16BPC222-C | 2200 | " | " | " |
| M123A16BPB272-C | 2700 | " | " | 50 |
| M123A16BPB332-C | 3300 | " | " | " |
| M123A16BPB392-C | 3900 | " | " | " |
| M123A16BPB472-C | 4700 | " | " | " |
| M123A16BXD271KC | 270 | K | BX | 200 |
| M123A16BXD331-C | 330 | K,M | " | " |
| M123A16BXD391KC | 390 | K | " | " |
| M123A16BXD471-C | 470 | K,M | " | " |
| M123A16BXD561KC | 560 | K | " | " |

See footnote at end of table.



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TABLE I. Style CKS22 characteristics - Continued.

| Part number 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (V dc) |
|---|---|---|---|---|
| M123A16BXD681-C | 680 | K,M | BX | 200 |
| M123A16BXD821KC | 820 | K | " | " |
| M123A16BXC102-C | 1000 | K,M | " | 100 |
| M123A16BXC122KC | 1200 | K | " | " |
| M123A16BXC152-C | 1500 | K,M | " | " |
| M123A16BXC182KC | 1800 | K | " | " |
| M123A16BXC222-C | 2200 | K,M | " | " |
| M123A16BXC272KC | 2700 | K | " | " |
| M123A16BXC332-C | 3300 | K,M | " | " |
| M123A16BXC392KC | 3900 | K | " | " |
| M123A16BXC472-C | 4700 | K,M | " | " |
| M123A16BXC562KC | 5600 | K | " | " |
| M123A16BXC682-C | 6800 | K,M | " | " |
| M123A16BXC822KC | 8200 | K | " | " |
| M123A16BXC103-C | 10000 | K,M | " | " |
| M123A16BXB123KC | 12000 | K | " | 50 |
| M123A16BXB153-C | 15000 | K,M | " | " |
| M123A16BXB183KC | 18000 | K | " | " |
| M123A16BXB223-C | 22000 | K,M | " | " |
| M123A16BXB273KC | 27000 | K | " | " |
| M123A16BXB333-C | 33000 | K,M | " | " |
| M123A16BXB393KC | 39000 | K | " | " |
| M123A16BXB473-C | 47000 | K,M | " | " |
| M123A16BXB563KC | 56000 | K | " | " |
| M123A16BXB683-C | 68000 | K,M | " | " |
| M123A16BXB823KC | 82000 | K | " | " |
| M123A16BXB104-C | 100000 | K,M | " | " |

1/ The complete part number shall include a symbol to indicate capacitance tolerance, as applicable.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Molded plastic encapsulated.

Lead wire: Type C of MIL-C-123.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature: -55°C to +125°C.

Voltage rating: See table I.

Temperature coefficient tolerances for BP values below 20 pF: In accordance with MIL-C-123.

Pre-encapsulation terminal strength: In accordance with MIL-C-123.

Marking: In accordance with MIL-C-123, example 6.



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## CONCLUDING MATERIAL

Custodians:
Air Force - 19
NASA - NA

Review activities:
Army - ER
Navy - EC
Air Force - 85
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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